

Fundamentals Physics

Eleventh Edition

Halliday



Chapter 3

Vectors

3-1 Vectors and Their Components (1 of 15)

Learning Objectives

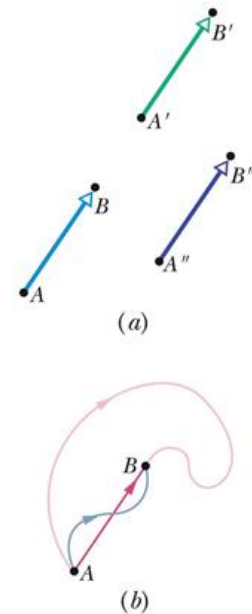
- 3.01** Add vectors by drawing them in head-to-tail arrangements, applying the commutative and associative laws.
- 3.02** Subtract a vector from a second one.
- 3.03** Calculate the components of a vector on a given coordinate system, showing them in a drawing.
- 3.04** Given the components of a vector, draw the vector and determine its magnitude and orientation.
- 3.05** Convert angle measures between degrees and radians.

3-1 Vectors and Their Components (2 of 15)

- Physics deals with quantities that have both size and direction
- A **vector** is a mathematical object with size and direction
- A **vector quantity** is a quantity that can be represented by a vector
 - Examples: position, velocity, acceleration
 - Vectors have their own rules for manipulation
- A **scalar** is a quantity that does not have a direction
 - Examples: time, temperature, energy, mass
 - Scalars are manipulated with ordinary algebra

3-1 Vectors and Their Components (3 of 15)

- The simplest example is a **displacement vector**
- If a particle changes position from A to B , we represent this by a vector arrow pointing from A to B
- In (a) we see that all three arrows have the same magnitude and direction: they are identical displacement vectors.
- In (b) we see that all three paths correspond to the same displacement vector. The vector tells us nothing about the actual path that was taken between A and B .



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Figure 3-1

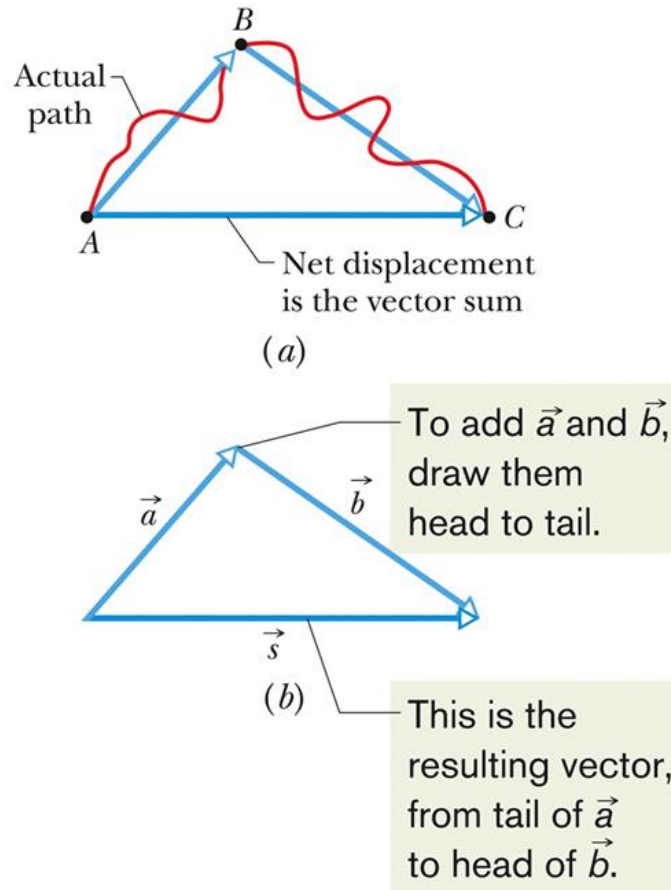
3-1 Vectors and Their Components (4 of 15)

- The **vector sum**, or **resultant**
 - Is the result of performing vector addition
 - Represents the net displacement of two or more displacement vectors

$$\vec{s} = \vec{a} + \vec{b}, \quad \text{Equation (3-1)}$$

- Can be added graphically as shown:

3-1 Vectors and Their Components (5 of 15)



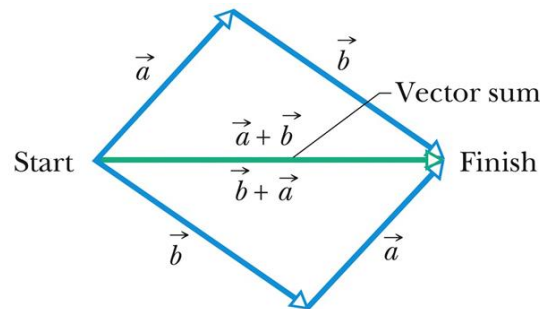
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Figure 3-2

3-1 Vectors and Their Components (6 of 15)

- Vector addition is **commutative**
 - We can add vectors in any order

$$\vec{a} + \vec{b} = \vec{b} + \vec{a} \quad (\text{commutative law}). \quad \text{Equation (3-2)}$$



You get the same vector result for either order of adding vectors.

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Figure (3-3)

3-1 Vectors and Their Components (7 of 15)

- Vector addition is **associative**
 - We can group vector addition however we like

$$(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c}) \quad (\text{associative law}). \quad \text{Equation (3-3)}$$

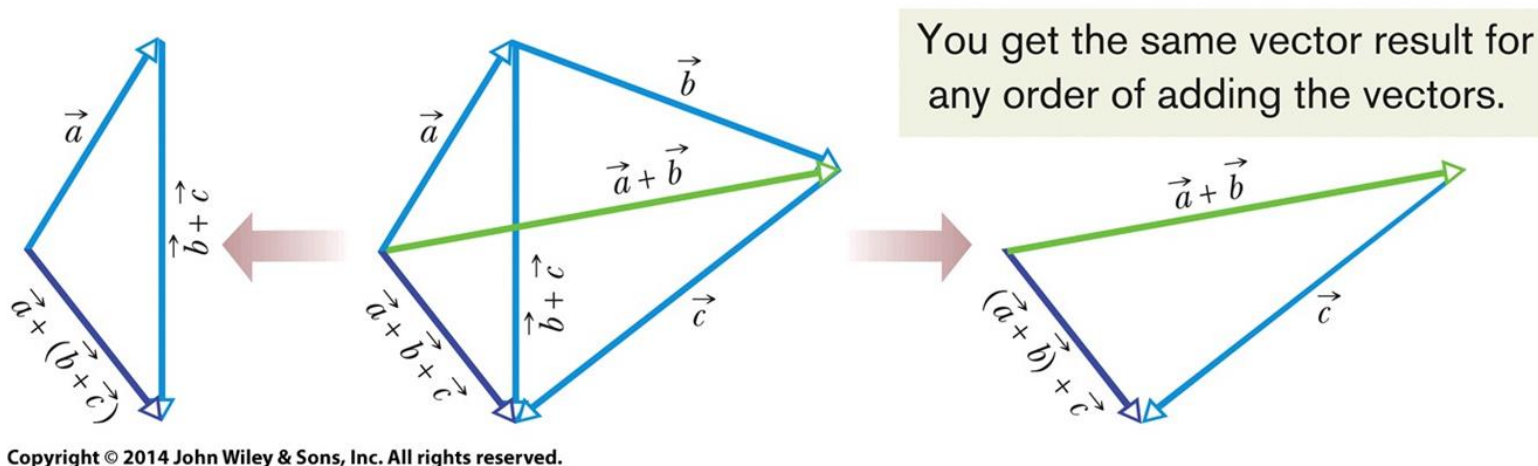


Figure (3-4)

3-1 Vectors and Their Components (8 of 15)

- A negative sign reverses vector direction

$$\vec{b} + (-\vec{b}) = 0.$$

- We use this to define vector subtraction

$$\vec{d} = \vec{a} - \vec{b} = \vec{a} + (-\vec{b})$$

Equation (3-4)

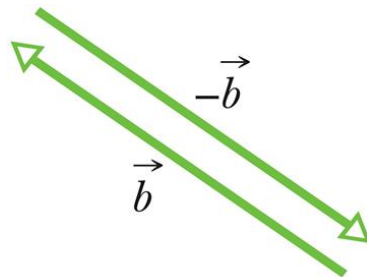


Figure (3-5)

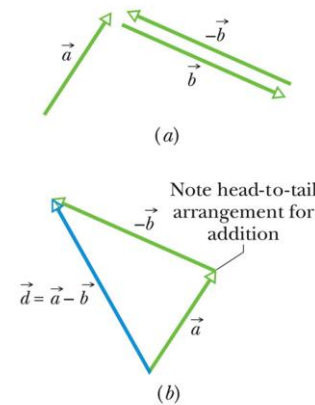


Figure (3-6)

3-1 Vectors and Their Components (9 of 15)

- These rules hold for all vectors, whether they represent displacement, velocity, etc.
- Only vectors of the same kind can be added
 - (distance) + (distance) makes sense
 - (distance) + (velocity) does not

3-1 Vectors and Their Components (10 of 15)

Checkpoint 1

The magnitudes of displacements $\vec{a}$ and $\vec{b}$ are 3 m and 4 m, respectively, and $\vec{c} = \vec{a} + \vec{b}$. Considering various orientations of $\vec{a}$ and $\vec{b}$, what are (a) the maximum possible magnitude for $\vec{c}$ and (b) the minimum possible magnitude?

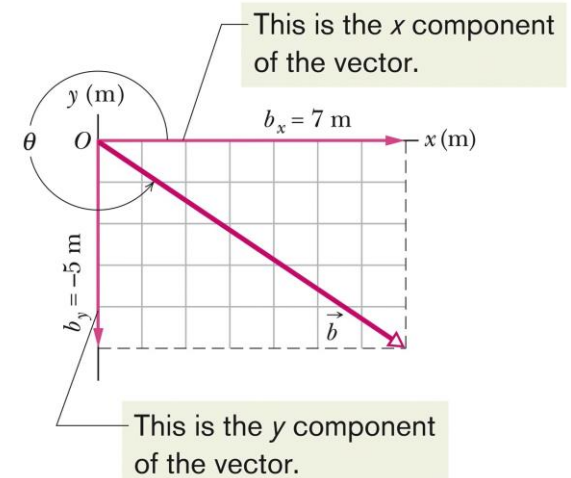
Answer:

(a) $3 \text{ m} + 4 \text{ m} = 7 \text{ m}$

(b) $4 \text{ m} - 3 \text{ m} = 1 \text{ m}$

3-1 Vectors and Their Components (11 of 15)

- Rather than using a graphical method, vectors can be added by **components**
 - A component is the projection of a vector on an axis
- The process of finding components is called **resolving the vector**
- The components of a vector can be positive or negative.
- They are unchanged if the vector is shifted in any direction (but not rotated).



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Figure (3-8)

3-1 Vectors and Their Components (12 of 15)

- Components in two dimensions can be found by:

$$a_x = a \cos \theta \quad \text{and} \quad a_y = a \sin \theta, \quad \text{Equation (3-5)}$$

- Where θ is the angle the vector makes with the positive x axis, and a is the vector length
- The length and angle can also be found if the components are known

$$a = \sqrt{a_x^2 + a_y^2} \quad \text{and} \quad \tan \theta = \frac{a_y}{a_x} \quad \text{Equation (3-6)}$$

- Therefore, components fully define a vector

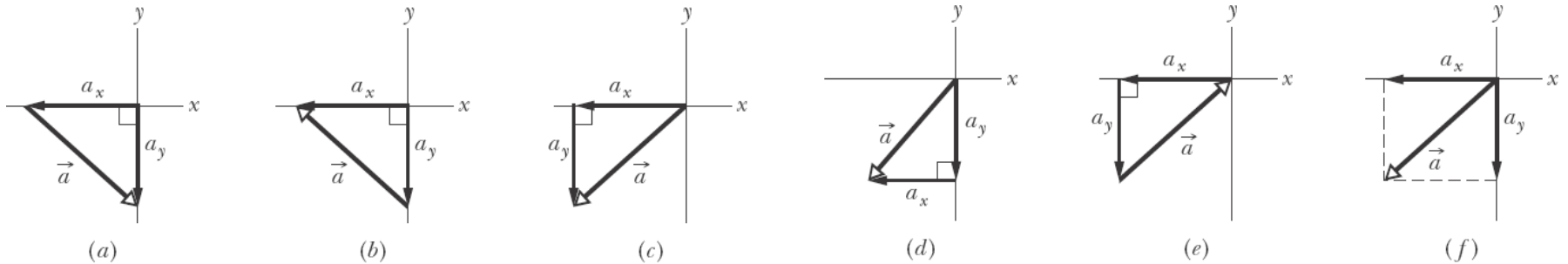
3-1 Vectors and Their Components (13 of 15)

- In the three dimensional case we need more components to specify a vector
 - (a, θ, ϕ) or (a_x, a_y, a_z)

3-1 Vectors and Their Components (14 of 15)

Checkpoint 2

In the figure, which of the indicated methods for combining the x and y components of vector $\vec{a}$ are proper to determine that vector?



Answer: choices (c), (d), and (f) show the components properly arranged to form the vector

3-1 Vectors and Their Components (15 of 15)

- Angles may be measured in degrees or radians
- Recall that a full circle is 360° , or 2π rad

$$40^\circ \frac{2\pi \text{ rad}}{360^\circ} = 0.70 \text{ rad.}$$

- Know the three basic trigonometric functions

$$\sin \theta = \frac{\text{leg opposite } \theta}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{leg adjacent to } \theta}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{leg opposite } \theta}{\text{leg adjacent to } \theta}$$

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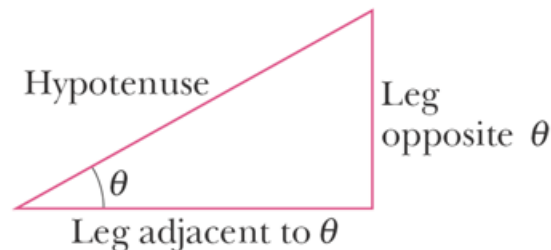


Figure (3-11)

3-2 Unit Vectors, Adding Vectors by Components (1 of 8)

Learning Objectives

- 3.06** Convert a vector between magnitude-angle and unit-vector notations.
- 3.07** Add and subtract vectors in magnitude-angle notation and in unit-vector notation.
- 3.08** Identify that, for a given vector, rotating the coordinate system about the origin can change the vector's components, but not the vector itself.

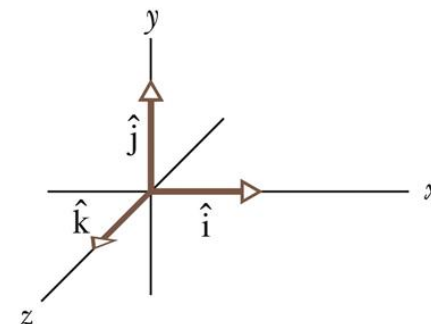
3-2 Unit Vectors, Adding Vectors by Components (2 of 8)

- **A unit vector**
 - Has magnitude 1
 - Has a particular direction
 - Lacks both dimension and unit
 - Is labeled with a hat: $\hat{}$
- We use a **right-handed coordinate system**
 - Remains right-handed when rotated

$$\vec{a} = a_x \hat{i} + a_y \hat{j} \quad \text{Equation (3-7)}$$

$$\vec{b} = b_x \hat{i} + b_y \hat{j} \quad \text{Equation (3-8)}$$

The unit vectors point along axes.



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Figure (3-13)

3-2 Unit Vectors, Adding Vectors by Components (3 of 8)

- The quantities $a_x\hat{i}$ and $a_y\hat{j}$ are **vector components**

$$\vec{a} = a_x\hat{i} + a_y\hat{j} \quad \text{Equation (3-7)}$$

$$\vec{b} = b_x\hat{i} + b_y\hat{j} \quad \text{Equation (3-8)}$$

- The quantities a_x and a_y alone are **scalar components**
 - Or just “components” as before

3-2 Unit Vectors, Adding Vectors by Components (4 of 8)

- Vectors can be added using components

Equation (3-9) $\vec{r} = \vec{a} + \vec{b}, \rightarrow r_x = a_x + b_x$

Equation (3-10)

$$r_y = a_y + b_y$$

Equation (3-11)

$$r_z = a_z + b_z.$$

Equation (3-12)

3-2 Unit Vectors, Adding Vectors by Components (5 of 8)

- To subtract two vectors, we subtract components

$$d_x = a_x - b_x, \quad d_y = a_y - b_y, \quad \text{and} \quad d_z = a_z - b_z,$$

$$\vec{d} = d_x \hat{i} + d_y \hat{j} + d_z \hat{k}. \quad \text{Equation (3-13)}$$

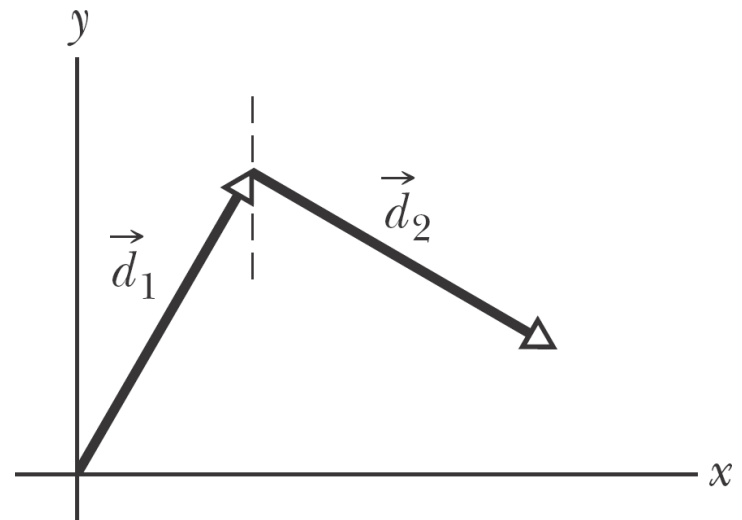
3-2 Unit Vectors, Adding Vectors by Components (6 of 8)

Checkpoint 3

(a) In the figure here, what are the signs of the x components of $\vec{d}_1$ and $\vec{d}_2$? (b) What are the signs of the y components of $\vec{d}_1$ and $\vec{d}_2$? (c) What are the signs of the x and y components of $\vec{d}_1 + \vec{d}_2$?

Answer:

- (a) positive, positive
- (b) positive, negative
- (c) positive, positive



3-2 Unit Vectors, Adding Vectors by Components (7 of 8)

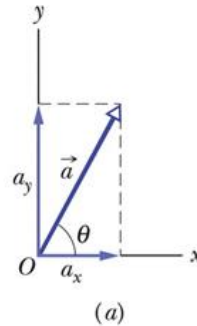
- Vectors are independent of the coordinate system used to measure them
- We can rotate the coordinate system, without rotating the vector, and the vector remains the same

$$a = \sqrt{a_x^2 + a_y^2} = \sqrt{a_x'^2 + a_y'^2} \quad \text{Equation (3-14)}$$

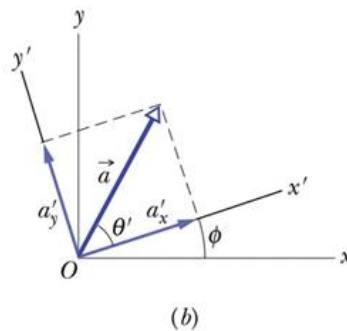
$$\theta = \theta' + \phi. \quad \text{Equation (3-15)}$$

- All such coordinate systems are equally valid

3-2 Unit Vectors, Adding Vectors by Components (8 of 8)



Rotating the axes changes the components but not the vector.



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Figure (3-15)

3-3 Multiplying Vectors (1 of 13)

Learning Objectives

- 3.09** Multiply vectors by scalars.
- 3.10** Identify that multiplying a vector by a scalar gives a vector, the dot product gives a scalar, and the cross product gives a perpendicular vector.
- 3.11** Find the dot product of two vectors.
- 3.12** Find the angle between two vectors by taking their dot product.
- 3.13** Given two vectors, use the dot product to find out how much of one vector lies along the other.

3-3 Multiplying Vectors (2 of 13)

- 3.14 Find the cross product of two vectors.
- 3.15 Use the right-hand rule to find the direction of the resultant vector.
- 3.16 In nested products, start with the innermost product and work outward.

3-3 Multiplying Vectors (3 of 13)

- Multiplying a vector $\mathbf{z}$ by a scalar c
 - Results in a new vector
 - Its magnitude is the magnitude of vector $\mathbf{z}$ times $|c|$
 - Its direction is the same as vector $\mathbf{z}$, or opposite if c is negative
 - To achieve this, we can simply multiply each of the components of vector $\mathbf{z}$ by c
- To divide a vector by a scalar we multiply by $\frac{1}{c}$

3-3 Multiplying Vectors (4 of 13)

Example Multiply vector $\mathbf{z}$ by 5

- $\mathbf{z} = -3\mathbf{i} + 5\mathbf{j}$
- $5\mathbf{z} = -15\mathbf{i} + 25\mathbf{j}$

3-3 Multiplying Vectors (5 of 13)

- Multiplying two vectors: the **scalar product**
 - Also called the **dot product**
 - Results in a scalar, where a and b are magnitudes and ϕ is the angle between the directions of the two vectors:

$$\vec{a} \cdot \vec{b} = ab \cos \phi, \quad \text{Equation (3-20)}$$

- The commutative law applies, and we can do the dot product in component form

$$\vec{a} \cdot \vec{b} = (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \cdot (b_x \hat{i} + b_y \hat{j} + b_z \hat{k}), \quad \text{Equation (3-22)}$$

$$\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}. \quad \vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z. \quad \text{Equation (3-23)}$$

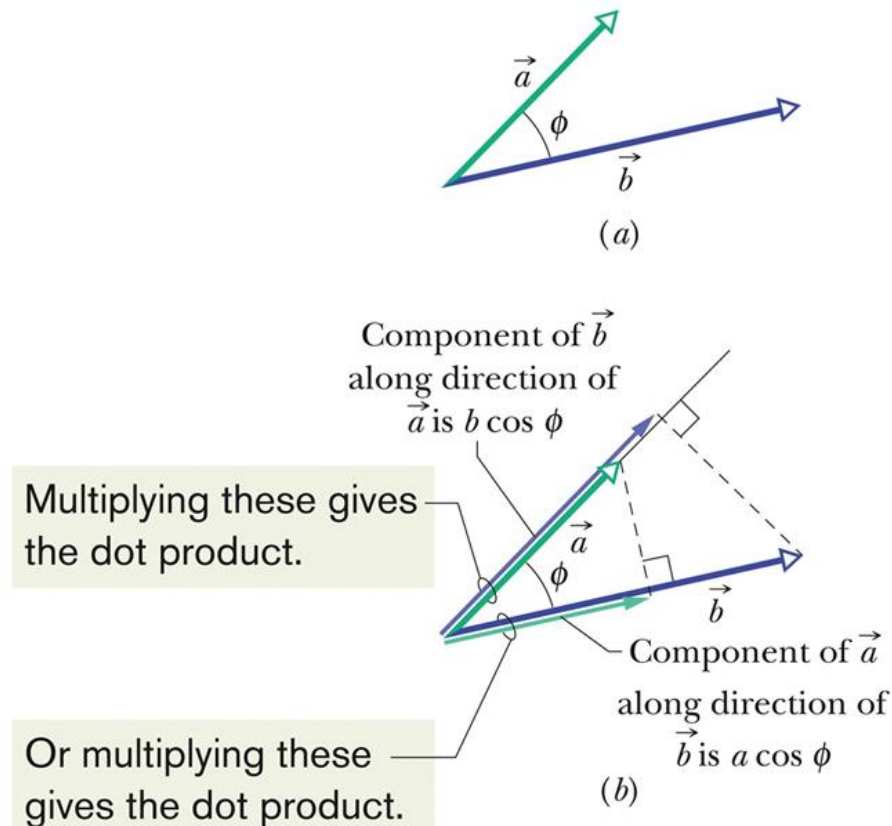
3-3 Multiplying Vectors (6 of 13)

- A dot product is: the product of the magnitude of one vector times the scalar component of the other vector in the direction of the first vector

$$\vec{a} \cdot \vec{b} = (a \cos \phi)(b) = (a)(b \cos \phi). \quad \text{Equation (3-21)}$$

- Either projection of one vector onto the other can be used
- To multiply a vector by the projection, multiply the magnitudes

3-3 Multiplying Vectors (7 of 13)



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Figure (3-18)

3-3 Multiplying Vectors (8 of 13)

If the angle ϕ between two vectors is 0° , the component of one vector along the other is maximum, and so also is the dot product of the vectors. If, instead, ϕ is 90° , the component of one vector along the other is zero, and so is the dot product.

Checkpoint 4

Vectors $\vec{C}$ and $\vec{D}$ have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of $\vec{C}$ and $\vec{D}$ if $\vec{C} \cdot \vec{D}$ equals (a) zero, (b) 12 units, and (c) -12 units?

Answer:

- (a) 90 degrees
- (b) 0 degrees
- (c) 180 degrees

3-3 Multiplying Vectors (9 of 13)

- Multiplying two vectors: the **vector product**
 - The **cross product** of two vectors with magnitudes a & b , separated by angle ϕ , produces a vector with magnitude:

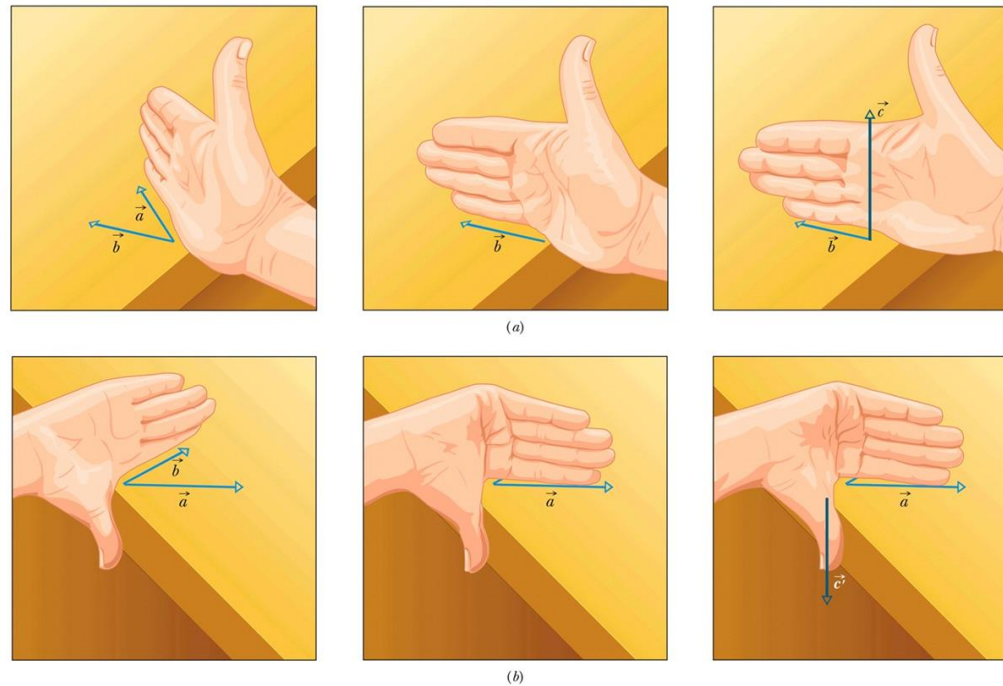
$$c = ab \sin \phi, \quad \text{Equation (3-24)}$$

- And a direction perpendicular to both original vectors
- Direction is determined by the **right-hand rule**
- Place vectors tail-to-tail, sweep fingers from the first to the second, and thumb points in the direction of the resultant vector

3-3 Multiplying Vectors (10 of 13)

If $\vec{a}$ and $\vec{b}$ are parallel or antiparallel, $\vec{a} \times \vec{b} = 0$. The magnitude of $\vec{a} \times \vec{b}$, which can be written as $|\vec{a} \times \vec{b}|$, is maximum when $\vec{a}$ and $\vec{b}$ are perpendicular to each other.

3-3 Multiplying Vectors (11 of 13)



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Figure (3-19)

The upper shows vector a cross vector b , the lower shows vector b cross vector a

3-3 Multiplying Vectors (12 of 13)

- The cross product is not commutative

$$\vec{b} \times \vec{a} = -(\vec{a} \times \vec{b}). \quad \text{Equation (3-25)}$$

- To evaluate, we distribute over components:

$$\vec{a} \times \vec{b} = (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \times (b_x \hat{i} + b_y \hat{j} + b_z \hat{k}), \quad \text{Equation (3-26)}$$

$$a_x \hat{i} \times b_x \hat{i} = a_x b_x (\hat{i} \times \hat{i}) = 0,$$

$$a_x \hat{i} \times b_y \hat{j} = a_x b_y (\hat{i} \times \hat{j}) = a_x b_y \hat{k}.$$

- Therefore, by expanding (3-26):

$$\vec{a} \times \vec{b} = (a_y b_z - b_y a_z) \hat{i} + (a_z b_x - b_z a_x) \hat{j} + (a_x b_y - b_x a_y) \hat{k}. \quad \text{Equation (3-27)}$$

3-3 Multiplying Vectors (13 of 13)

Checkpoint 5

Vectors $\vec{C}$ and $\vec{D}$ have magnitudes of 3 units and 4 units, respectively. What is the angle between the directions of $\vec{C}$ and $\vec{D}$ if the magnitude of the vector product $\vec{C} \times \vec{D}$ is (a) zero and (b) 12 units?

Answer:

(a) 0 degrees

(b) 90 degrees

3 Summary (1 of 4)

Scalars and Vectors

- Scalars have magnitude only
- Vectors have magnitude and direction
- Both have units!

Vector Components

- Given by

$$a_x = a \cos \theta \quad \text{and} \quad a_y = a \sin \theta, \quad \text{Equation (3-5)}$$

- Related back by

$$a = \sqrt{a_x^2 + a_y^2} \quad \text{and} \quad \tan \theta = \frac{a_y}{a_x} \quad \text{Equation (3-6)}$$

3 Summary (2 of 4)

Adding Geometrically

- Obeys commutative and associative laws

$$\vec{a} + \vec{b} = \vec{b} + \vec{a}$$

Equation (3-2)

$$(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c}).$$

Equation (3-3)

Unit Vector Notation

- We can write vectors in terms of unit vectors

$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k},$$

Equation (3-7)

3 Summary (3 of 4)

Adding by Components

- Add component-by-component

$$r_x = a_x + b_x$$

$$r_y = a_y + b_y$$

$$r_z = a_z + b_z.$$

Equations (3-10) - (3-12)

Scalar Product

- Dot product

$$\vec{a} \cdot \vec{b} = ab \cos \phi,$$

Equation (3-20)

$$\vec{a} \cdot \vec{b} = (a_x \hat{i} + a_y \hat{j} + a_z \hat{k}) \cdot (b_x \hat{i} + b_y \hat{j} + b_z \hat{k}),$$

Equation (3-22)

3 Summary (4 of 4)

Scalar Times a Vector

- Product is a new vector
- Magnitude is multiplied by scalar
- Direction is same or opposite

Cross Product

- Produces a new vector in perpendicular direction
- Direction determined by right-hand rule

$$c = ab \sin \phi, \quad \text{Equation (3-24)}$$